Logistic Regression and Classification

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

For the matrix $A = (1, 2, 0, 1; 2, 4, 1, 4; 3, 6, 1, 5)$, find bases for $\text{Row}(A)$, $\text{Col}(A)$, and $\text{Null}(A)$. Verify the Rank–Nullity Theorem.

2 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A random sample of 200 students is classified by Grade Level (9th vs 12th) and Smartphone Operating System (iOS vs Android). Observed counts: 9th Grade: 70 iOS, 30 Android (100 total); 12th Grade: 50 iOS, 50 Android (100 total). (a) Calculate the expected count of 9th graders with iOS. (b) Compute the Chi-square test statistic χ^2. (c) With $df = 1$ and $\alpha = 0.05$ (critical value 3.84), conclude whether Grade Level and Operating System are independent.

3 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Perform a complete bifurcation analysis of the system in terms of parameter $k \in \mathbb{R}$: $\{x + y + kz = 2; 3x + 4y + 2z = k; 2x + 3y - z = 1$ Determine all values of k for which the system has (a) a unique solution, (b) infinitely many solutions, and (c) no solution.

4 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Verify the Cayley–Hamilton Theorem for $A = (2, 1; 3, 4)$, and use it to compute A^{-1} without using cofactor formulas or row reduction.

5 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

6 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Let $A = (1, 2, 0; 0, 3, 1; 2, 1, 4)$. Compute $\det(A)$, find the adjugate matrix $\text{adj}(A)$, and verify the identity $A\text{adj}(A) = \det(A)I_3$.

7 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t\rangle$ for $0 \le t \le 2\pi$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

9 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Prove that the limit $\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

10 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

11 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

12 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

In an SRS of 200 registered voters, 116 favor a new environmental regulation. (a) Check all three conditions for constructing a 95% confidence interval for the population proportion. (b) Construct the 95% confidence interval. (c) Interpret the interval in context.

13 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right) dx$.

14 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Level 2 · MediumExplain why the matrix $A = (3, 1; 0, 3)$ is not diagonalizable.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Level 3 · Advanced / University RigorLet A be a 5×8 matrix with $\text{rank}(A) = 4$. (a) What is the dimension of the solution space to $A\mathbf{x} = \mathbf{0}$? (b) Is $A\mathbf{x} = \mathbf{b}$ consistent for every $\mathbf{b} \in \mathbb{R}^5$? Explain. (c) What is the dimension of $\text{Null}(A^T)$?

16 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

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17 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x, y, z) = (x + 2y - z,\ 2x + 5y,\ 3x + 7y - z)$. Find bases for $\ker(T)$ and $\text{im}(T)$, and determine whether T is an isomorphism.

18 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

19 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

20 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. maximum

d. negative

22 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0

b. K

c. ∞

d. -1

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23 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In a bivariate regression with sample size $n = 31$ (Research Model 1), sample slope is $b_1 = 1.70$ with $\text{SE}_{b_1} = 0.26$. Testing $H_0 : \beta_1 = 0$, what is the t-test statistic and degrees of freedom?

a. $t = 6.54$, df $= 29$

b. $t = 0.44$, df $= 29$

c. $t = 8.08$, df $= 31$

d. $t = 6.54$, df $= 30$

24 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

In a regression analysis with $n = 23$ data points, the slope estimate is $b_1 = 2.31$ with standard error $SE_{b_1} = 0.909$. Find the test statistic t for testing $H_0 : \beta_1 = 0$.

a. 1.04

b. 4.54

c. 1.54

d. 2.54

25 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

A regression line predicts $\hat{y} = 41.85$ for $x = 25$, while the observed value is $y = 39.1$. What is the residual for this point?

a. 3.25

b. -2.75

c. -0.75

d. -6.75

26 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Inflection point classification: $f''(c) = 0$ is necessary but not sufficient. Example of $f''(0) = 0$ with no inflection:

a. x^4

b. x^3

c. x^2

d. $\sin x$

27 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic equation $P' = rP(1 - P/K)$ has equilibria at:

a. $P = 0$ only

b. $P = K$ only

c. $P = r$

d. $P = 0$ and $P = K$

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28 [Foundational] • Target: Logistic Regression & Classification Practice Exercises

Logistic equation steady states (equilibria):

a. $P = 0$ only

b. $P = K$ only

c. $P = 0$ and $P = K$

d. $P = K/2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

The least-squares regression line for a data set is $\hat{y} = 11.1 + 2.5x$. What is the predicted value of y when $x = 45$?

a. 128.6

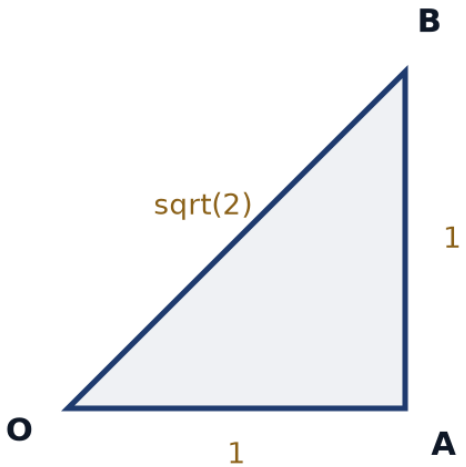
b. 123.6

c. 118.6

d. 125.6

30 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.



a. 1/12

b. 1/8

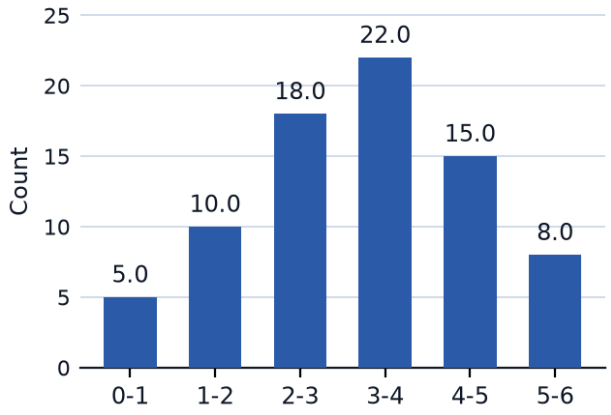
c. 1/6

d. 1/4

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31 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

The histogram below displays the distribution of the number of hours students spent studying for a recent exam. What is the approximate median number of hours studied?



a. 2.5 hours

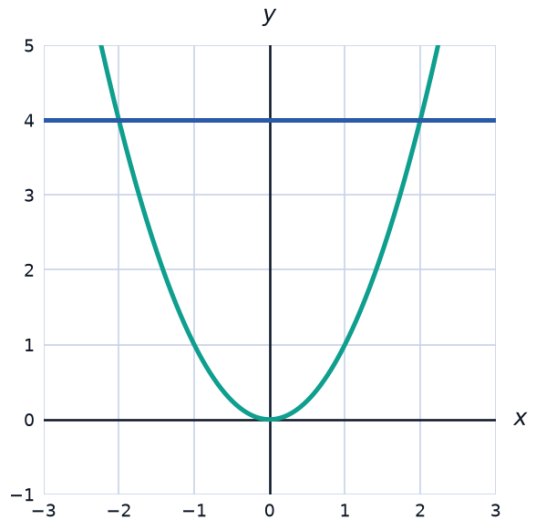
b. 3.5 hours

c. 4.5 hours

d. 5.5 hours

32 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?



a. $\frac{32}{3}$

b. $\frac{16}{3}$

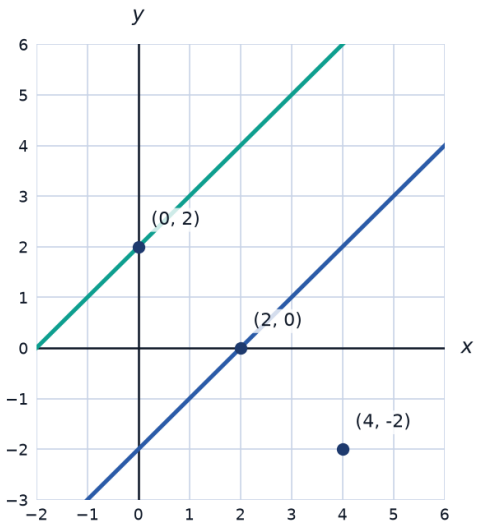
c. 8

d. 16

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33 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?



a. 17

b. 7

c. 2.4

d. -7

34 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1, 5)$

b. $[-1, 5]$

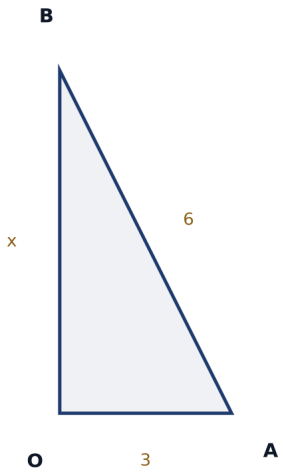
c. $(-1, 5]$

d. $[-1, 5)$

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35 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\, dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta + 2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\, dr\, d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \leq \theta \leq \frac{\pi}{2}$.



a. $\alpha = 0, \beta = \frac{\pi}{2}$

b. $\alpha = 0, \beta = \frac{\pi}{4}$

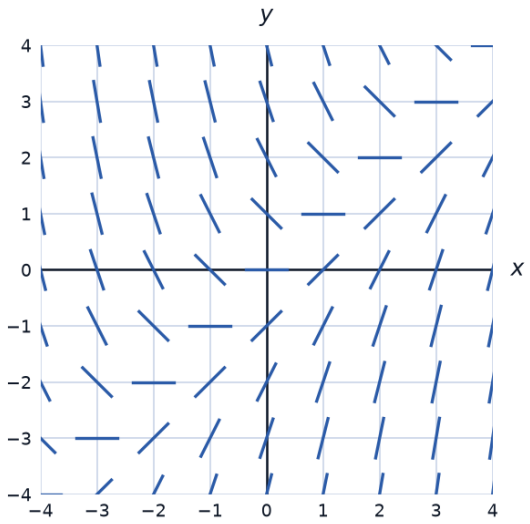
c. $\alpha = 0, \beta = \frac{3\pi}{4}$

d. $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

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36 **[Intermediate]** • **Target: Logistic Regression & Classification Practice Exercises**

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$



a. $\frac{dy}{dx} = \frac{x}{y}$

b. $\frac{dy}{dx} = \frac{y}{x}$

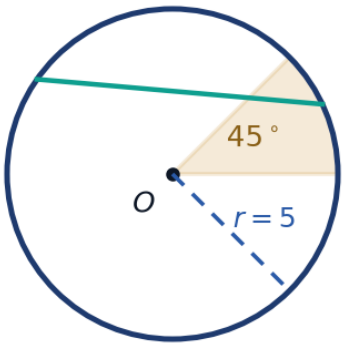
c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

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37 [Intermediate] • Target: Logistic Regression & Classification Practice Exercises

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.



a. 4

b. $\pi/4$

c. 16

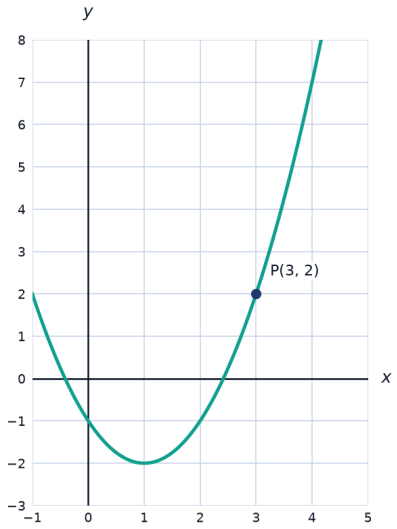
d. 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



a. 4

b. 2

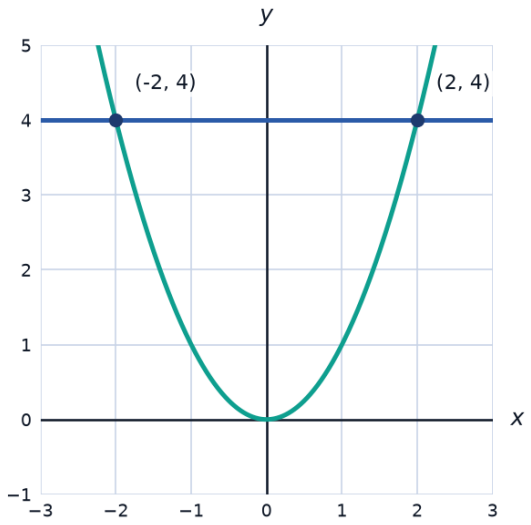
c. 3

d. 6

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39 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi \int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi \int_{-2}^{2} ((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi \int_{0}^{4} ((4 - x^2)^2 - r^2)dx$

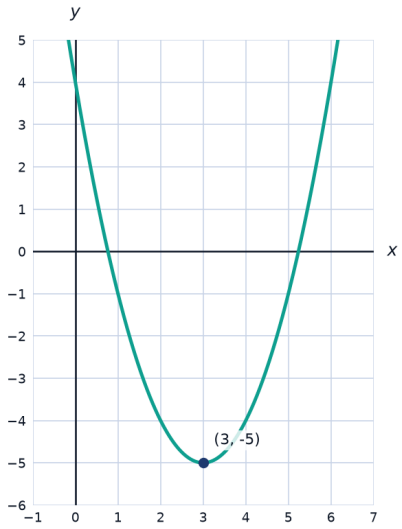
c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi \int_{1}^{4} ((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi \int_{-2}^{2} ((4 - x^2)^2 - r^2)dx$

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40 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.



a. -5

b. 4

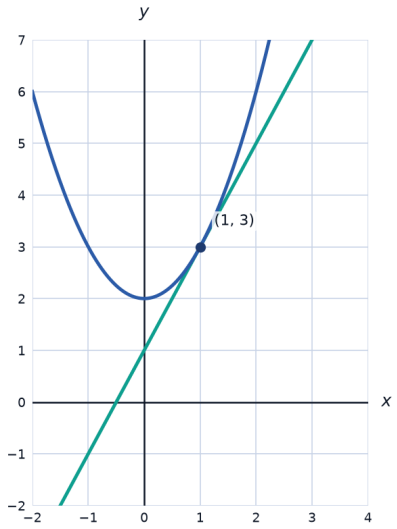
c. 9

d. 0

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41 **[Challenging]** • **Target: Logistic Regression & Classification Practice Exercises** • **Cross-Lesson: 6.01 Linear Regression and OLS**

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.



a. 1

b. 2

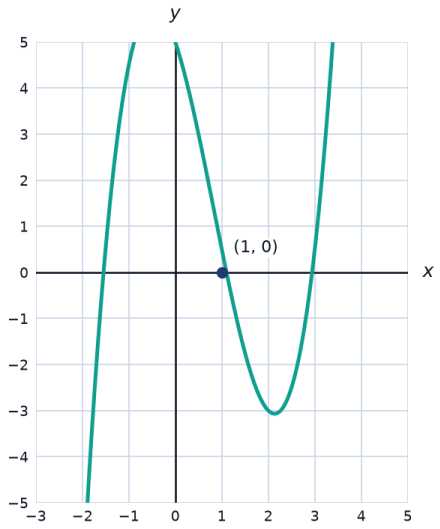
c. 3

d. 4

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42 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?



a. $(-2, 3)$

b. $(0, -1)$

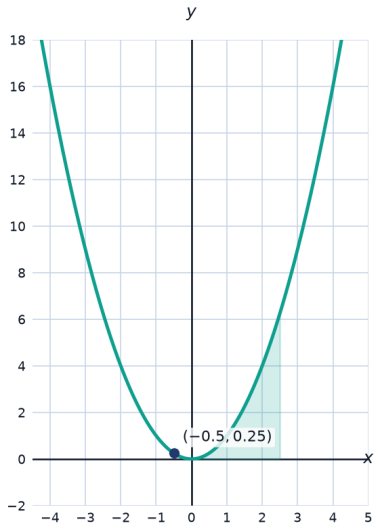
c. $(1, 0)$

d. $(3, 4)$

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43 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

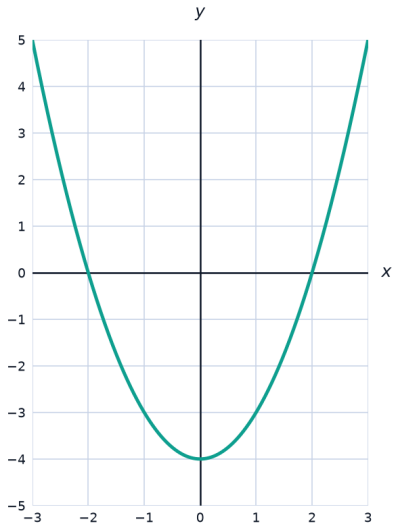
c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

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44 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.



a. 2

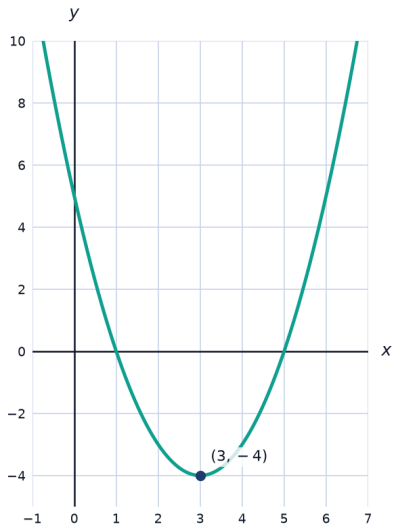
b. 4

c. 8

d. 16

45 [Challenging] • Target: Logistic Regression & Classification Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the function $f(x) = x^2 - 6x + 5$. The function has a local minimum at which value of x?



a. 0

b. 3

c. 6

d. 5

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